

One Hundred and Fifty Publication-Quality Figures

A practical guide, with the complete worked-example library

Yanwei Wang

companion to the `publication-figures` repository

(scripts `fig001–fig150`, `publication.mplstyle`, `journal_style.py`, `ashby.py`, `manuscript.py`)

Contents

1	The craft: how these figures are made	1
1.1	One style sheet for the whole paper	1
1.2	Derive, don't draw	1
1.3	Geometry and typography	1
1.4	Colour and encoding	2
1.5	Annotation beats legends	2
1.6	Multi-panel discipline	2
1.7	Verification is part of the figure	2
1.8	Reproducibility and packaging	2
2	The catalog: one hundred and fifty worked examples	2
3	The failure-mode catalog	19
4	Checklists	20

1 The craft: how these figures are made

Every figure in this library was produced the same way, and the method is the point. The scripts are worked examples of eight habits.

1.1 One style sheet for the whole paper

All scripts load a single `publication.mplstyle`: single-column figures about 3.5–4.0in wide (double column 7.0–7.2in), 7–8pt text so labels match journal captions after scaling, the Okabe–Ito colour-blind-safe cycle, despined axes with outward ticks (the `journal_style` module switches to boxed inward ticks when imitating older journals), `constrained_layout` for spacing, and twin outputs: a 600-dpi PNG for previewing and a vector PDF for submission. Consistency is not cosmetic — reviewers read a paper whose figures share one visual grammar far faster.

1.2 Derive, don't draw

No curve in this library is sketched. Each is computed from the governing equations — Nernst lines, Butler–Volmer kinetics, the Buckley–Leverett fractional flow, Planck's law — with generic textbook parameters. When the task is to *reproduce* a figure from a paper (fig. 61), the workflow is: read every label and axis; derive the underlying solution; verify landmark values the original states; parameterise the computation; add asymptotic expansions the original implies; imitate the styling (“style archaeology”); iterate against the target. A reproduced figure you derived is one you understand — and one you may legally redraw, because the physics is nobody's property while the original's pixels are.

1.3 Geometry and typography

Work in inches and points, never pixels: set the final printed size and let DPI follow. Label every axis with symbol *and* unit. Prefer `mathtext` for symbols and remember it is a dialect of $\text{\LaTeX}$, not $\text{\LaTeX}$ itself (pattern 14). Type μ , $^\circ$ and the minus sign as literal characters — escape sequences inside raw strings print themselves (pattern 1).

1.4 Colour and encoding

Default to a colour-blind-safe cycle and never let colour carry meaning alone: pair it with line style or marker shape, and check the figure in grayscale. Reserve semantic colour (red for failure envelopes, warm for hot isotherms) for when it teaches.

1.5 Annotation beats legends

With few curves, label them directly, in their own colour, where the curves are *separated* — converging families (sag curves, titration tails) must be labelled in their fan, not at their confluence (pattern 11). Rotated labels along sloped lines must use the rendered axes aspect, not the data slope (pattern 2); `ashby.slope_angle` automates this. Keep annotations off the data; use thin leaders when the target region is crowded; aim leaders at the named object's ink, not the nearest convenient coordinate (pattern 15). Bottom-edge labels collide with tick numbers — place them inside the axes (pattern 10).

1.6 Multi-panel discipline

One message per panel; bold sequential panel letters; shared axes where comparison is the point; `height_ratios` for sketch strips and residual panels; insets never over data — relocate or promote to a panel (pattern 3).

1.7 Verification is part of the figure

Run, *look at the pixels*, fix, rerun — plausible code routinely produces wrong figures, and the render is the only witness. Build self-checks into the physics wherever possible: the Wien locus must thread every Planck peak (fig. 93); Mohr circles must touch the envelope because centre-to-line distance equals radius (fig. 85); the Carnot loop integral must equal the T - s rectangle, and an `assert` says so (fig. 100). Seed every random generator; print confirmations and key derived numbers on save.

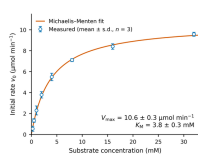
1.8 Reproducibility and packaging

One standalone script per figure with a marked DATA block; shared code promoted to modules (`journal_style.py`, `ashby.py`, `manuscript.py`); a README indexing every script in one line; the whole library shipped as a versioned zip. A figure others can regenerate is a figure others can trust.

2 The catalog: one hundred and fifty worked examples

Each entry shows the figure, what it demonstrates, and (where the description names one) the built-in check it carries. Thumbnails are downsampled; run the script for the full-resolution PNG and vector PDF. Sections A–H are the original hundred; section I adds twenty manuscript-style panels built at exact printed size, and section J thirty figure types from further fields on the same exact-size module.

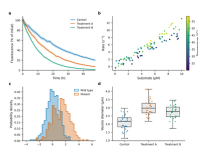
A • Statistical and general data graphics (figs. 1–50)



1. Data + model fit

Data + model fit with error bars (`curve_fit`, parameter annotation)

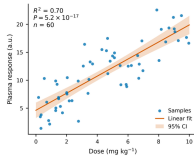
Techniques: annotation



2. Multipanel

2×2 composite with bold a–d panel labels, s.e.m. bands, colorbar, distributions, box + raw points

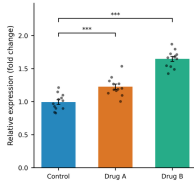
Techniques: colorbar, direct labels



3. Regression ci

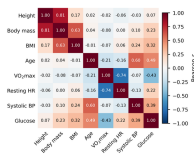
Scatter + linear regression with an analytically computed 95% confidence band and $R^2/P/n$ annotation

Techniques: annotation



4. Bars stats

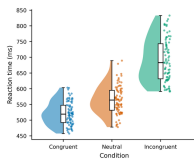
Bars with individual data points, s.e.m. error bars, and significance brackets from t-tests



5. Heatmap

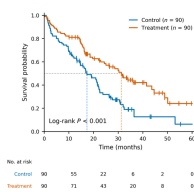
Annotated correlation heatmap with a diverging colormap pinned to [-1, 1]

Techniques: annotation



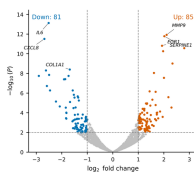
6. Raincloud

Raincloud plots: half-violin distribution + box summary + jittered raw data



7. Kaplan–Meier survival

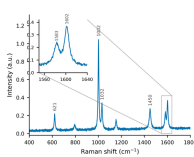
Kaplan–Meier survival curves with censor marks, median guides, log-rank test ($\text{scipy} \geq 1.11$), and a number-at-risk table



8. Volcano

Volcano plot with thresholds, per-side labeled top hits, and rasterized points for small PDFs

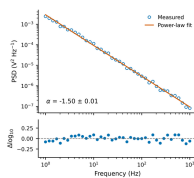
Techniques: rasterized points, direct labels



9. Spectrum inset

Annotated spectrum with a zoom inset via `inset_axes` + `indicate_inset_zoom`

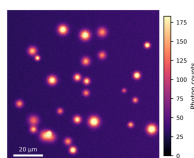
Techniques: inset axes, annotation



10. Loglog powerlaw

Log-log power-law fit with a residual panel (`height_ratios` + `sharex`)

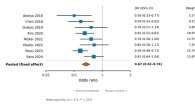
Techniques: log axes



11. Image scalebar

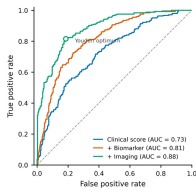
Microscopy-style image: scale bar, percentile contrast stretch, calibrated colorbar, no axis ticks

Techniques: colorbar



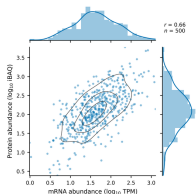
12. Forest plot

Meta-analysis forest plot: weighted squares, pooled diamond, aligned OR/weight text columns, heterogeneity stats



13. ROC curves

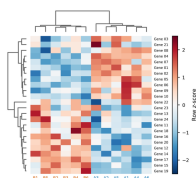
ROC curves with AUC computed from scratch in numpy and a marked Youden optimum



14. Joint marginals

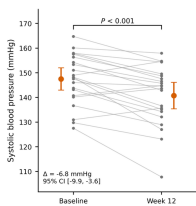
Scatter with marginal histograms/KDEs via GridSpec (a "joint plot" without seaborn)

Techniques: GridSpec



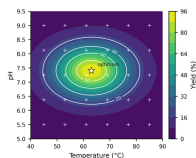
15. Clustered heatmap

Clustered heatmap with scipy dendrograms aligned to the reordered matrix (a "clustermap" from scratch)



16. Paired slope

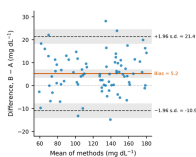
Paired before/after slope chart with per-subject lines, group means $\pm$ 95% CI, and paired t-test



17. Contour map

2-D response surface: filled + labelled contours, design points, marked optimum

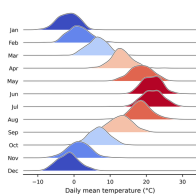
Techniques: direct labels, shading, contours



18. Bland–Altman comparison

Bland–Altman method comparison: bias, limits of agreement, CI bands, right-edge value labels

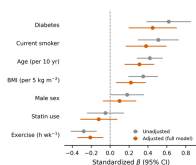
Techniques: direct labels



19. Ridgeline

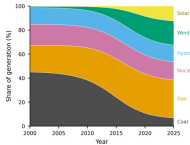
Ridgeline (joy) plot: overlapping KDEs with occlusion and fills colour-mapped to each row's mean

Techniques: shading



20. Coefficients

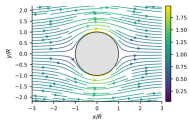
Regression coefficient dot-and-whisker plot comparing adjusted vs unadjusted models



21. Stacked composition

Normalized stacked-area composition with semantic colours and direct right-edge labels (no legend)

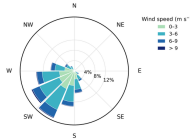
Techniques: legend craft, direct labels



22. Streamplot

Vector field as speed-coloured streamlines with a masked body and an inset colorbar matched to equal-aspect axes

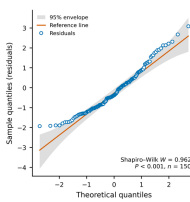
Techniques: inset axes, colorbar, aspect-aware rotation



23. Wind rose

Wind rose: stacked speed classes on a compass-oriented polar axis, radial percent ticks

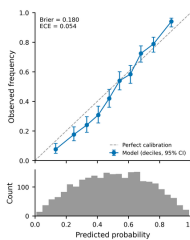
Techniques: polar axes



24. Normal Q-Q plot

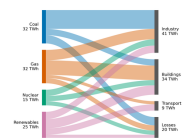
Normal Q-Q plot with quartile reference line, Beta-distribution 95% envelope, and Shapiro-Wilk test

Techniques: envelopes



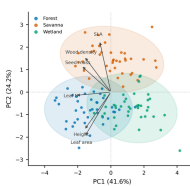
25. Calibration

Calibration curve with Wilson CIs, Brier/ECE, and a prediction-histogram panel



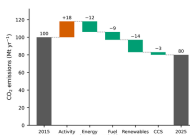
26. Sankey

Sankey/alluvial flow diagram from scratch: smoothstep ribbons via `fil_between`, widths $\propto$ flow



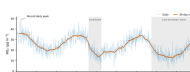
27. PCA biplot

PCA via SVD with 95% confidence ellipses per group and loading arrows



28. Waterfall

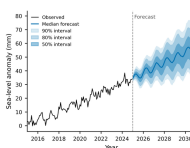
Waterfall chart: floating signed contributions between start and end totals with connectors



29. Time series events

Dated time series: ConciseDateFormat axis, rolling mean, shaded event periods, annotated peak

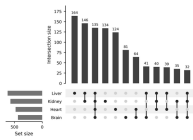
Techniques: series solution, annotation, shading



30. Fan chart

Forecast fan chart: nested 50/80/90% quantile bands from an ensemble with proxy-patch legend

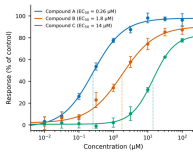
Techniques: legend craft



31. UpSet plot

UpSet plot of set intersections: bars + dot matrix + set sizes, GridSpec with shared axes

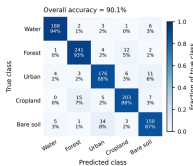
Techniques: GridSpec



32. Dose response

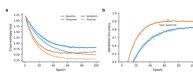
Dose-response curves: 4-parameter logistic fits in log space, EC50 drops and labels

Techniques: direct labels



33. Confusion matrix

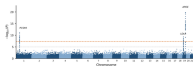
Row-normalized confusion matrix with counts, percentages, and adaptive text colour



34. Training curves

ML training curves: mean $\pm$ s.d. across seeds, train/val line styles, best-epoch marker

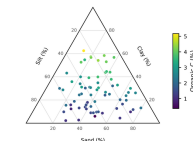
Techniques: seeded RNG



35. Manhattan plot (GWAS)

GWAS Manhattan plot: alternating chromosome colours, significance thresholds, rasterized points, labelled loci

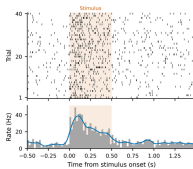
Techniques: rasterized points, direct labels



36. Ternary

Ternary composition diagram from scratch: grid, edge ticks, rotated axis titles, inset colorbar

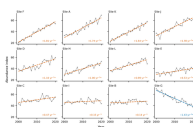
Techniques: inset axes, colorbar



37. Spike raster + PSTH

Spike raster (eventplot) + PSTH with Gaussian-smoothed rate and shaded stimulus window

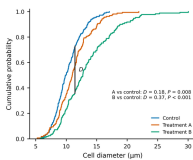
Techniques: shading



38. Small multiples

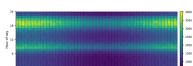
Small-multiples grid: shared axes, per-panel trend fits coloured by sign, sorted panels, supxlabel/supylabel

Techniques: direct labels



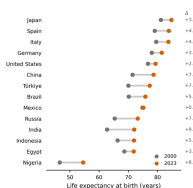
39. ECDF comparison

ECDF comparison via Axes.ecdf with two-sample KS tests and the maximum-gap marker



40. Carpet plot

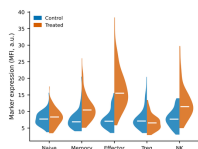
Hour-of-day $\times$ day-of-year carpet plot with pcolormesh cell edges and month-centred ticks



41. Dumbbell

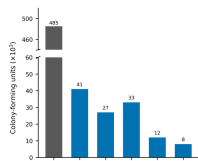
Dumbbell chart of change between two time points, sorted, with Δ labels at the right edge

Techniques: direct labels



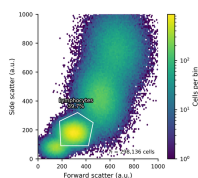
42. Split violin

Split violins: two conditions per category sharing a baseline, white median bars



43. Broken axis

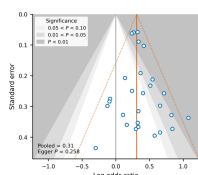
Bar chart with a broken y-axis: stacked axes, hidden joint, diagonal break marks



44. Hexbin density

Hexbin density with LogNorm for $10^5 +$ points, polygon gate with enclosed fraction, stroked labels

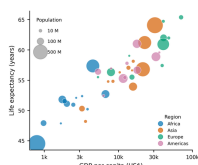
Techniques: direct labels



45. Funnel plot

Contour-enhanced funnel plot: significance shading, pseudo-95% funnel, Egger's test

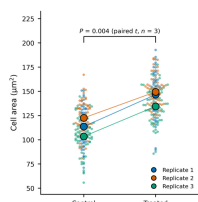
Techniques: shading, contours



46. Bubble chart

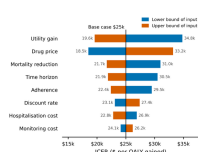
Bubble chart: log axis with plain ticks, colour by group, hand-built size legend (diameter = $\sqrt{\text{area}}$)

Techniques: legend craft



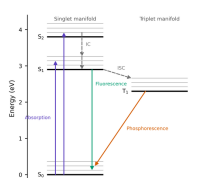
47. Superplot

SuperPlot: from-scratch beeswarm coloured by replicate, replicate means overlaid, test on $n =$ replicates



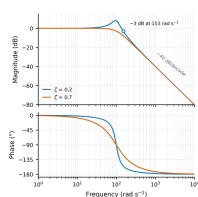
48. Tornado

Tornado diagram for one-way sensitivity analysis, sorted by swing, currency tick formatter



49. Energy levels

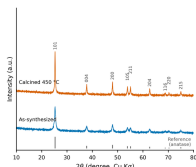
Jablonski energy-level diagram drawn in code: states, vibrational sublevels, radiative/non-radiative arrows



50. Bode plot

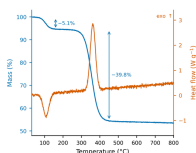
Bode magnitude/phase plot from scipy.signal with -3 dB marker, slope guide, log grid

B • Chemistry and materials characterization (figs. 51–60)



51. X-ray diffraction

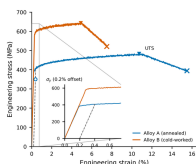
Stacked powder XRD with reference stick pattern and rotated (hkl) indices



52. TGA–DSC thermogram

TGA + DSC on a colour-coded twin axis with mass-loss arrows and "exo up" marker

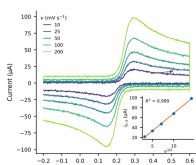
Techniques: twin axes



53. Stress–strain curves

Engineering stress–strain: 0.2% offset yield construction, UTS/fracture markers, elastic-region inset

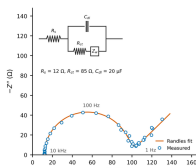
Techniques: inset axes



54. Cyclic voltammetry

CVs at five scan rates with a Randles–Ševčík (i_p vs $\sqrt{v}$) inset

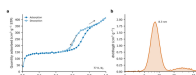
Techniques: inset axes



55. Nyquist EIS

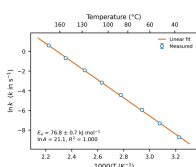
Nyquist impedance plot with Randles-model fit, frequency tags, and a code-drawn equivalent-circuit inset

Techniques: inset axes



56. Isotherm + PSD

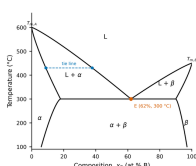
Type-IV N₂ isotherm with H1 hysteresis (IUPAC open/closed symbols) + BJH pore-size panel



57. Arrhenius plot

Arrhenius plot with weighted fit, E_a , and an exact secondary °C axis via **secondary_axis**

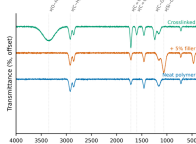
Techniques: secondary axis



58. Phase diagram

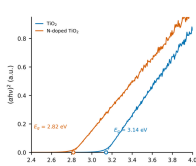
Parameterised binary eutectic phase diagram: labelled fields, eutectic point, tie line

Techniques: direct labels



59. FTIR stack

Stacked FTIR on a reversed wavenumber axis with dotted band guides and rotated assignments

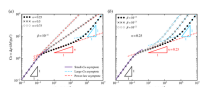


60. Tauc band gap

Tauc plot: linear-region fits extrapolated to the abscissa for optical band gaps

C • Case study: reproducing a journal theory figure (fig. 61)

This single figure carries the reproduction workflow of §I.2 and introduced `journal_style.py` (boxed inward ticks, slope triangles, panel letters), reused by every figure after it.



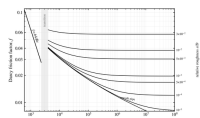
61. Carreau flow curves (case study)

RHEOLOGY

Journal-style theory figure: Carreau slit-flow curves (Cu vs Q) computed from the wall-shear integral, with small- Cu /power-law/large- Cu asymptotes, slope triangles, marker-triple legends, boxed inward-tick axes, and STIX serif math

Techniques: legend craft

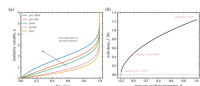
D • Transport phenomena and reactors (figs. 62–67)



62. Moody chart

Moody diagram computed from $64/Re$ and the Colebrook equation (vectorised fixed-point iteration), transition band, direct roughness labels

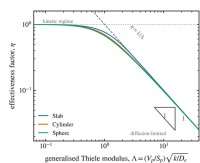
Techniques: direct labels



63. Falkner–Skan profiles

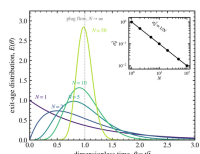
Falkner–Skan boundary layers by shooting (`solve_ivp` + `brentq` with blow-up events): profiles across the pressure-gradient range and wall shear $f''(0)$ vs β with separation/Blasius/stagnation anchors

Techniques: root finding, ODE integration



64. Thiele modulus

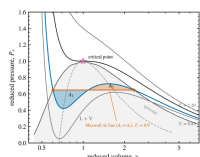
Effectiveness factor vs generalised Thiele modulus for slab/cylinder/sphere ($\tanh$, Bessel I_1/I_0 , $\coth$), asymptotes and slope triangle



65. RTD tanks-in-series

Tanks-in-series residence-time distributions with the $\sigma^2=1/N$ scaling inset

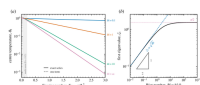
Techniques: inset axes, series solution



66. Van der Waals + Maxwell

Van der Waals isotherms with the Maxwell equal-area construction solved by `brentq` (analytic antiderivative), binodal dome, analytic spinodal, shaded equal-area lobes

Techniques: root finding, shading

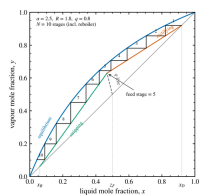


67. Transient conduction

Slab-conduction eigenseries ($\zeta \tan \zeta = Bi$ solved per branch), one-term Heisler approximation with validity band, and ζ_1 vs Bi with both asymptotes

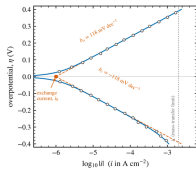
Techniques: series solution

E • Chemical and materials engineering classics (figs. 68–73)



68. McCabe–Thiele construction

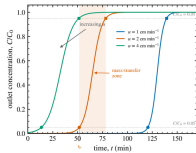
McCabe–Thiele construction with the stage-stepping algorithm coded (equilibrium/operating lines, sloped q -line, numbered stages, automatic feed-stage detection)



69. Tafel plot

Tafel plot from Butler–Volmer kinetics with a cathodic mass-transfer limit, dashed extrapolations meeting at i_0 , slope labels matched to the axes aspect

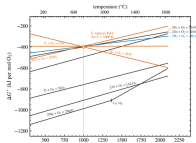
Techniques: direct labels, aspect-aware rotation



70. Breakthrough curves

Fixed-bed breakthrough curves (Thomas-type logistic), breakthrough/exhaustion markers, shaded mass-transfer zone

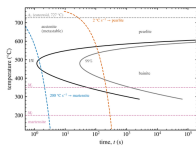
Techniques: shading



71. Ellingham diagram

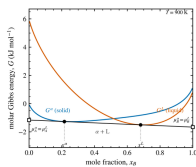
Ellingham diagram from $\Delta H - T\Delta S$ with the Mg boiling-point kink, the exact 1000 K carbothermic Fe crossing, along-line labels plus nudged right-margin labels with leaders

Techniques: direct labels



72. TTT diagram

TTT diagram whose C-curves emerge from $\ln t = a/T + b/(T(T_m - T)^2)$, with $A_1/Ms/Mf$ lines and two cooling paths (quench $\rightarrow$ martensite, slow $\rightarrow$ pearlite)

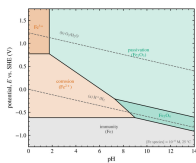


73. Common-tangent construction

Common-tangent construction on regular-solution G curves, solved robustly by parameterising the tangent by slope (brentq), with chemical-potential intercepts and coexisting compositions

Techniques: root finding, tangency construction

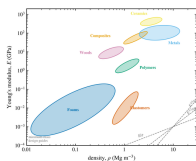
F • Electrochemistry, stability and selection (figs. 74–79)



74. Pourbaix diagram

Simplified Fe Pourbaix (E – pH) diagram: every boundary a Nernst line, self-consistent triple points, water-stability window, filled immunity/corrosion/passivation regions

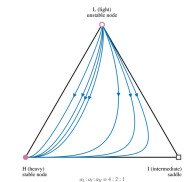
Techniques: shading



75. Ashby chart: E–rho

Ashby modulus–density selection chart: material-family ellipses drawn parametrically in log space, minimum-mass guide lines E/ρ , $E^{1/2}/\rho$, $E^{1/3}/\rho$ with aspect-matched labels

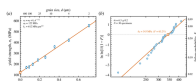
Techniques: direct labels, aspect-aware rotation



76. Residue curve map

Ternary residue curve map integrated from $dx/d\xi = x - y$ (constant α), node topology labelled, direction arrows anchored mid-descent to dodge the near-node crawl

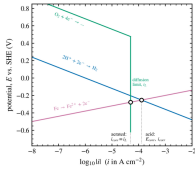
Techniques: direct labels



77. Hall–Petch + Weibull

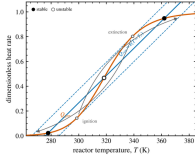
Hall–Petch fit with a nonlinear secondary grain-size axis, plus Weibull median-rank analysis with the double-log linearisation and a secondary failure-probability axis

Techniques: secondary axis



78. Evans diagram

Evans mixed-potential diagram: anodic dissolution vs activation- and diffusion-controlled cathodics; $i_{corr} = i_L$ when the anodic line meets the vertical limit



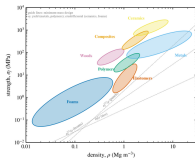
79. CSTR multiplicity

CSTR van Heerden diagram: sigmoid heat generation vs removal lines, three steady states (stable/unstable), ignition and extinction tangencies found by brentq on $x'(T) = \kappa$

Techniques: root finding

G • Ashby charts on the shared module (figs. 80–81)

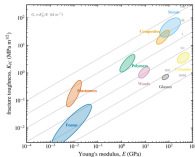
Both charts are thin scripts over `ashby.py`: log-space bubbles with the major axis along the tilt, decade axes, and guide-line labels rotated from the *rendered* axes aspect — the module that retired failure pattern 2.



80. Ashby chart: strength–density

Ashby strength–density chart on the module, with minimum-mass guide lines for ties (σ/ρ), beams ($\sigma^{2/3}/\rho$) and panels ($\sigma^{1/2}/\rho$)

Techniques: shared module

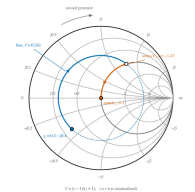


81. Ashby chart: toughness–modulus

Ashby K_{IC} – E chart with toughness contours $G_c = K^2/E$ (slope $\frac{1}{2}$), making the square in the toughness relation visible

Techniques: contours

H • Across science and engineering (figs. 82–100)

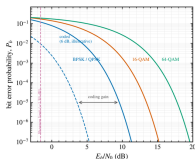


82. Smith chart

RF

Smith chart generated from the bilinear map $\Gamma = (z-1)/(z+1)$, with a worked two-step match: a series line ($\ell \approx 0.24\lambda$, clockwise at constant $-\Gamma$) onto $r = 1$, then a series capacitor to the centre

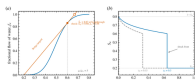
Techniques: series solution



83. BER waterfalls

COMMUNICATIONS

BER waterfalls from Q-function expressions for BPSK/QPSK, 16-QAM and 64-QAM, an illustrative 6 dB coding-gain shift, and the Shannon limit at -1.59 dB

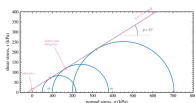


84. Buckley–Leverett waterflood

PETROLEUM

Buckley–Leverett waterflood: Welge tangent from $(S_{wc}, 0)$ fixing the shock saturation and breakthrough average, plus rarefaction-and-shock saturation profiles at two times

Techniques: tangency construction

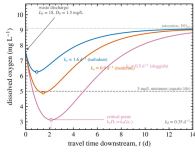


85. Mohr–Coulomb failure

GEOTECHNICAL

Mohr–Coulomb failure circles tangent to $\tau = c + \sigma \tan \phi$ by construction (centre-to-line distance equals radius analytically), with the friction angle and cohesion annotated at equal aspect

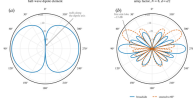
Techniques: annotation, tangency construction, aspect-aware rotation



86. Streeter–Phelps DO sag

ENVIRONMENTAL

Streeter–Phelps dissolved-oxygen sag for three reaeration rates, critical points from the analytic t_c and D_c , and the 5 mg/L aquatic-life threshold

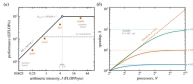


87. Antenna patterns

ANTENNAS

Polar dB patterns computed from array theory: half-wave dipole element factor and the $N = 8$, $d = \lambda/2$ array factor, broadside and steered to 60° ($\beta = -\pi \cos 60^\circ$), with the -13 dB first side lobe

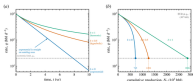
Techniques: polar axes



88. Roofline + Amdahl

COMPUTER ARCH.

Roofline model ($P = \min(P_{peak}, BW \cdot I)$), ridge point, no-vectorisation ceiling, kernels under their roofs; reuses `ashby.setup/slope_angle` beside Amdahl saturation vs Gustafson scaling

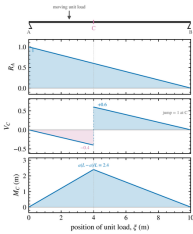


89. Arps decline curves

PETROLEUM

Arps decline family $q = q_i / (1 + b D_{it})^{1/b}$ on semilog time ($b = 0$ straight) and as rate–cumulative with analytic $N_p(q)$: EUR at the economic limit spans $715 \rightarrow 2856 \cdot 10^3$ bbl from $b = 0$ to 1

Techniques: log axes

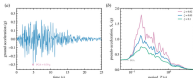


90. Influence lines

STRUCTURAL

Influence lines for R_A , V_C and M_C of a simply supported beam from statics, with beam sketch, positive/negative shading, the unit shear jump at C and the $a(L-a)/L$ moment peak

Techniques: shading

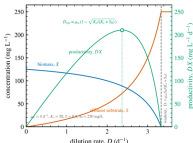


91. Response spectrum

EARTHQUAKE

Elastic response spectrum computed honestly: seeded band-passed, enveloped ground motion (PGA 0.30 g) integrated per-period with a vectorised central-difference SDOF sweep at $\zeta = 2, 5, 10 \%$; $S_a \rightarrow$ PGA at short period

Techniques: seeded RNG, envelopes

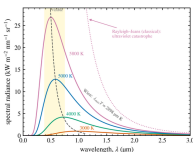


92. Chemostat washout

BIOPROCESS

Chemostat washout diagram from Monod kinetics: S^* , X^* , and productivity $D \cdot X$ vs dilution rate, washout at $D_c = \mu_m S_0 / (K_s + S_0)$, analytic D_{opt} asserted against the numerical maximum

Techniques: built-in self-check

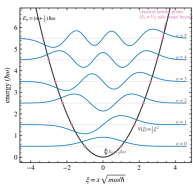


93. Planck blackbody

PHYSICS

Planck spectra at four temperatures with the Wien displacement locus drawn through the peaks (a built-in check), the Rayleigh–Jeans divergence, and the visible band

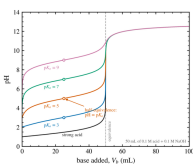
Techniques: built-in self-check



94. Quantum harmonic oscillator

QUANTUM

Quantum harmonic oscillator: Hermite–Gaussian eigenfunctions riding their levels $E_n = (n + \frac{1}{2}) \hbar \omega$ inside $V = \xi^2/2$, classical turning points dotted, zero-point energy bracketed at the well bottom

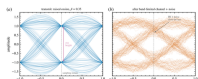


95. Titration curves

CHEMISTRY

Titration curves solved exactly from the proton condition with `brentq` (dilution included): strong + four weak acids, half-equivalence circles landing at $\text{pH} = \text{pKa}$ by construction

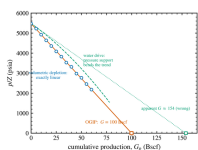
Techniques: root finding



96. Eye diagrams

COMMUNICATIONS

Eye diagrams from folded raised-cosine traces ($\beta = 0.35$, singularities handled analytically): the Nyquist pinch at ± 1 at the transmitter, then ISI + noise closing the eye after a band-limited channel

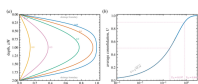


97. p/Z material balance

PETROLEUM

Gas material balance: noisy synthetic surveys on the exact p/Z line recover OGIP by extrapolation ($G_{hat} = 99.8$ vs true 100), with the water-drive bend and its misleading early-time tangent (apparent $G \approx 154$)

Techniques: tangency construction

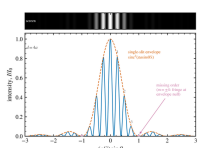


98. Terzaghi consolidation

GEOTECHNICAL

Terzaghi consolidation from the diffusion series: pore-pressure isochrones for a doubly drained layer and $U(T_v)$ with the $\sqrt{4T_v/\pi}$ approximation and the $T_{.50}/T_{.90}$ milestones

Techniques: series solution

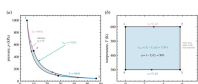


99. Double-slit interference

OPTICS

Double-slit interference under its single-slit envelope, with a rendered screen strip: $d = 4a$ makes the $m = \pm 4$ orders coincide with envelope nulls and vanish — the missing orders are the built-in check

Techniques: built-in self-check, envelopes



100. Carnot cycle, both planes

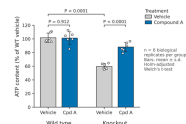
THERMODYNAMICS

The Carnot cycle in both planes: computed p-V branches enclosing the net work beside the T-s rectangle, with the loop integral asserted equal to $(T_h - T_c)\Delta S$ — 1729 J either way, because work between the same states through the same reversible cycle cannot care which plane you drew it in

Techniques: built-in self-check

I • Manuscript panels at exact size (figs. 101–120)

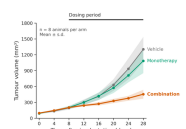
These twenty mostly revisit figure types from sections A and B as a journal prints them. Each is a script over `manuscript.py`: the canvas is set in millimetres and saved without a tight crop (89, 120 or 183 mm wide), type is 7/6 pt with a 5 pt floor, panel letters sit at a fixed point offset, and aligned plot areas are asserted. Every script also carries a built-in check of its own. All data are simulated.



101. Grouped bars, planned comparisons

Two-factor (genotype $\times$ treatment) bars with a second label row, every replicate over mean $\pm$ s.d., and exact Holm-adjusted Welch P values on brackets placed from the data maximum; the Holm step is asserted against its definition

Techniques: direct labels, built-in self-check

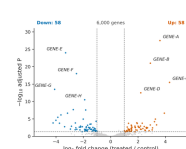


102. Growth curves by arm

ONCOLOGY

Per-animal longitudinal simulation summarised as mean $\pm$ s.d. bands, arms labelled at the right edge in their own colour instead of a legend, a dosing-period bar above the axes; pre-dosing arm means asserted to agree

Techniques: legend craft, direct labels, built-in self-check

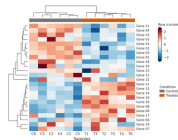


103. Volcano, BH-adjusted

TRANSCRIPTOMICS

Volcano on Benjamini–Hochberg adjusted P from per-gene standard errors, the BH step coded and asserted equal to scipy's, 6,000 genes as a rasterized layer under vector axes, class counts in a header row

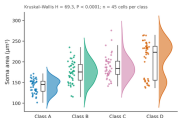
Techniques: rasterized points, built-in self-check



104. Annotated clustermap

Clustermap with correlation-distance average linkage on both axes, a condition bar between the column dendrogram and the matrix, colorbar and legend placed in mm; a two-cluster cut is asserted to reproduce the conditions

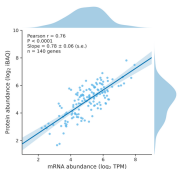
Techniques: colorbar, legend craft, built-in self-check



105. Raincloud + Kruskal–Wallis

Raincloud with KDE clouds trimmed where the density is negligible, a bimodal group its box alone would hide, and a Kruskal–Wallis header with n; KDE areas and mode counts asserted

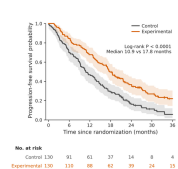
Techniques: built-in self-check



106. Regression + marginals

OLS line with an analytic 95% band for the mean response plus marginal KDEs on shared axes from a millimetre GridSpec; slope $\pm$ s.e., r, P and n in-panel; residual orthogonality and the band's waist at the mean of x asserted

Techniques: GridSpec, built-in self-check

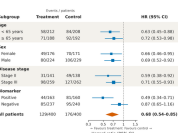


107. Kaplan–Meier with bands

CLINICAL TRIALS

Kaplan–Meier curves with 95% Greenwood log-log confidence bands, censor ticks and a number-at-risk table on the tick positions; the hand-coded log-rank test is asserted to reproduce scipy's

Techniques: log axes, built-in self-check

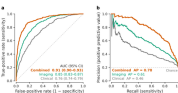


108. Subgroup forest plot

CLINICAL TRIALS

Trial subgroup forest: grouped rows with banding, events/patients text columns set in mm, weight-sized squares, overall diamond, plain-tick log hazard-ratio axis; arm totals asserted identical for every factor

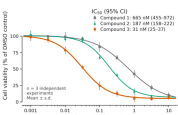
Techniques: built-in self-check



109. ROC + precision–recall

ROC beside precision–recall on equal square panels, AUC with 95% bootstrap intervals, direct labels in each empty corner; the rank-sum AUC is asserted equal to the trapezoid area under the plotted curve

Techniques: direct labels, built-in self-check

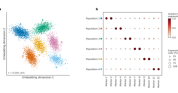


110. Dose–response with IC50 intervals

PHARMACOLOGY

4-parameter logistic fitted to all replicates with log IC50 as the parameter and a 95% CI from the covariance matrix; mean $\pm$ s.d. with distinct markers, plain decimal ticks; true IC50 asserted inside each interval

Techniques: built-in self-check

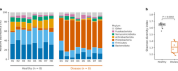


111. Embedding + marker dot plot

SINGLE-CELL

Single-cell style pair: a 6,000-point simulated embedding (rasterized layer, direct labels) beside a marker dot plot (area = % expressing, colour = scaled mean) with a hand-built size legend and mm-placed colorbar

Techniques: rasterized points, colorbar, legend craft, direct labels

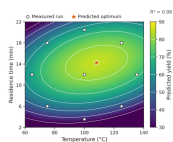


112. Composition + diversity

MICROBIOME

Per-sample stacked bars for two groups with group brackets and a stack-ordered legend, beside a Shannon-diversity box-plus-points panel with a Mann–Whitney test; every bar asserted to close at 100%

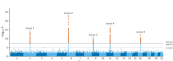
Techniques: legend craft, built-in self-check



113. Fitted response surface

Quadratic response surface fitted by least squares to a 13-run central composite design: optimum solved from the Hessian (asserted negative definite), R^2 , contour levels echoed on the colorbar

Techniques: colorbar, built-in self-check, contours

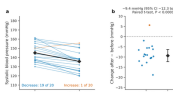


114. Manhattan plot, labelled loci

GENETICS

Manhattan plot of about 90,000 variants on real autosome proportions: rasterized points, thresholds named in the right margin, loci labelled at their lead variants; genomic inflation of the null asserted ≈ 1

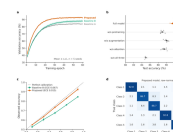
Techniques: rasterized points, direct labels, built-in self-check



115. Paired estimation plot

Paired slope chart coloured by direction of change, beside an estimation panel of the paired differences with their mean and 95% t interval; scipy's paired t asserted equal to the hand formula

Techniques: built-in self-check

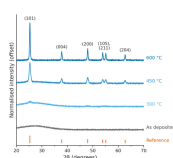


116. Model benchmark, four panels

MACHINE LEARNING

Four-panel ML benchmark as one argument on a mm grid: training curves, seed-level ablation dots with Δ versus the full model, reliability diagram with ECE, confusion matrix; mean $\pm$ s.d. over 5 seeds throughout

Techniques: seeded RNG

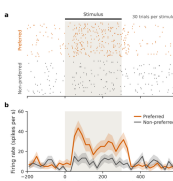


117. XRD annealing series

MATERIALS

Annealing series of pseudo-Voigt diffraction patterns sharpening out of an amorphous hump: sequential blues, right-margin trace labels, Miller indices at local maxima; peak finding asserted to recover all six reflections

Techniques: series solution, direct labels, built-in self-check

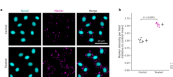


118. Raster + PSTH with s.e.m.

NEUROSCIENCE

Two-condition spike raster over a binned PSTH with mean $\pm$ s.e.m. across trials as a band, rotated block labels, stimulus bar, shared time axis; binning asserted to conserve every spike

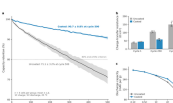
Techniques: direct labels, built-in self-check



119. Image plate + quantification

MICROSCOPY

2 $\times$ 3 dark image plate (nuclei, marker, merge) in cyan/magenta with shared linear display limits and one calibrated scale bar, beside field-level quantification measured from the same simulated images



120. Asymmetric hero composite

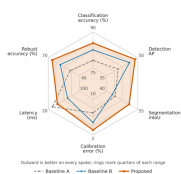
BATTERIES

Asymmetric composite: a hero time course spanning two rows beside two stacked supporting panels, one role per panel, mean $\pm$ s.d. ($n = 4$) throughout, direct labels; hero edges asserted to match the stack within 1.5 pt

Techniques: direct labels, built-in self-check

J • Beyond the standard chart: evidence panels across fields (figs. 121–150)

These thirty add figure types the first 120 did not cover: model evaluation (121–126), genomics (127–132), clinical research (133–138), networks, flows, sets and rankings (139–144), and signals, inference, dynamics and imaging (145–150). All are scripts over `manuscript.py` at exact printed size. Several code their layout algorithm from scratch — chord ribbons, a squarified treemap, the streamgraph baseline, a force-directed graph, sequence-logo glyphs — and each asserts a landmark the mathematics fixes. All data are simulated.

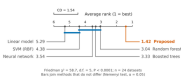


121. Radar benchmark comparison

MACHINE LEARNING

Radar chart of three methods on six tasks built on a bare polar axes: every spoke carries its own worst-to-best scale so outward is always better, proposed method filled, baselines outlined; shoelace polygon area asserted equal to its closed form and largest for the proposed method

Techniques: polar axes, built-in self-check, shading

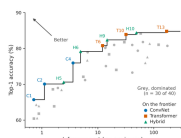


122. Critical-difference diagram

MACHINE LEARNING

Demšar critical-difference diagram for six classifiers on 24 datasets: average ranks on a reversed axis, Nemenyi CD ruler, thick bars joining methods that do not differ, names on elbow leaders; hand-coded Friedman χ^2 asserted equal to scipy's

Techniques: built-in self-check

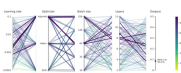


123. Pareto frontier

MACHINE LEARNING

Accuracy against inference cost (log axis, plain ticks) for 40 models in three families: Pareto frontier as a staircase, frontier models coloured and labelled directly, dominated models grey; sorted-sweep frontier asserted equal to the brute-force non-dominated set

Techniques: direct labels, built-in self-check

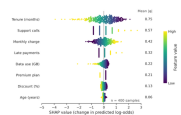


124. Parallel-coordinates sweep

MACHINE LEARNING

Parallel coordinates of a 150-run hyperparameter sweep over five axes in real units (log, categorical, power-of-two, linear), lines coloured by validation accuracy, top five runs thick, tick labels in clear lanes; every axis map asserted to round-trip and the labelled best run to be the argmax

Techniques: direct labels, built-in self-check

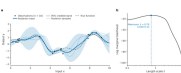


125. SHAP beeswarm summary

MACHINE LEARNING

SHAP summary beeswarm for a linear model, where the Shapley values are exact: 400 samples $\times$ 8 features sorted by mean absolute contribution, from-scratch swarm layout, dots coloured by feature value; additivity (contributions sum to prediction minus mean prediction) asserted for every sample

Techniques: built-in self-check

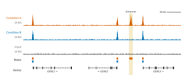


126. Gaussian-process regression

MACHINE LEARNING

Gaussian-process regression coded from the kernel equations (RBF kernel, Cholesky solve): posterior mean, 95% band and sample paths over noisy observations, beside the log marginal likelihood against length scale with its maximiser; Cholesky mean asserted equal to the direct solve

Techniques: built-in self-check



127. Genome browser tracks

GENOMICS

Genome-browser panel for a 200-kb window: three coverage tracks piled up from simulated fragments on one printed y scale, peak calls, gene models with strand chevrons, a shaded enhancer, Mb axis and scale bar; each coverage integral asserted equal to reads $\times$ fragment length

Techniques: built-in self-check, shading



128. Sequence logo

GENOMICS

Sequence logo of a 12-position motif from 200 simulated sites: letters are TextPath outlines stretched to exact boxes, stacked by information content with the small-sample correction, colour-blind-safe bases; glyph heights asserted to sum to 2 - H - e(n) bits

Techniques: built-in self-check

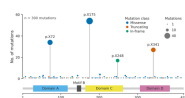


129. Oncoprint, memo-sorted

CANCER GENOMICS

Oncoprint of 12 genes $\times$ 60 samples: copy-number events fill the cell and mutations sit as an inner band so two can coexist, memo-sorted columns give the mutual-exclusivity staircase, with subtype row and marginal bars; printed percentages asserted equal to the row means

Techniques: built-in self-check, shading

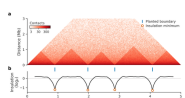


130. Mutation lollipop plot

CANCER GENOMICS

Protein lollipop plot: head area proportional to recurrence, colour by mutation class, over a domain bar labelled inside where the name fits; labelled hotspots asserted to be exactly the residues above a Poisson background threshold

Techniques: direct labels, built-in self-check

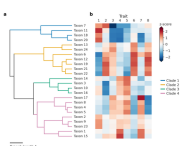


131. Hi-C contact triangle

3D GENOME

Simulated Hi-C map (power-law decay $\times$ domain enrichment) as a 45°-rotated rasterized triangle on a log colour scale, over an insulation-score track on the same genomic axis; every planted boundary asserted to have an insulation minimum within two bins

Techniques: rasterized points, built-in self-check

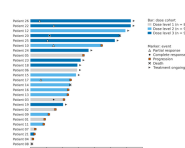


132. Tree with trait heatmap

PHYLOGENETICS

UPGMA tree of 24 taxa drawn from the linkage matrix with branches coloured by clade, aligned tip labels and a scale bar, beside a row-matched trait heatmap; tip order, ultrametric root-to-tip lengths and a cophenetic correlation above 0.8 asserted

Techniques: direct labels, built-in self-check

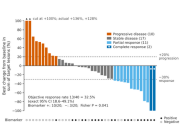


133. Swimmer plot

CLINICAL TRIALS

Swimmer plot of time on treatment for 26 patients: bars shaded by dose cohort and sorted by duration, marker shapes for response, progression and death, arrow heads for ongoing treatment; every event asserted to lie on its own bar

Techniques: built-in self-check, shading

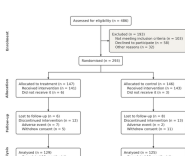


134. Response waterfall

ONCOLOGY

Waterfall of best tumour change for 40 patients with threshold lines, response categories derived from them, cut bars flagged, a biomarker strip, and the response rate with its exact 95% CI; Clopper–Pearson limits from beta quantiles asserted equal to scipy's

Techniques: built-in self-check

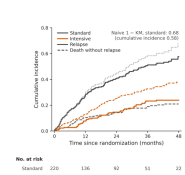


135. CONSORT flow diagram

CLINICAL TRIALS

CONSORT participant-flow diagram drawn in code: counts tallied from a simulated enrolment table, boxes sized from rendered text extents, right-angle connectors placed in millimetres; counts asserted to conserve at every node and every text to lie inside its box

Techniques: built-in self-check

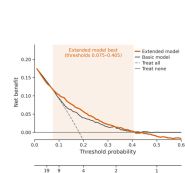


136. Competing-risks incidence

SURVIVAL ANALYSIS

Hand-coded Aalen–Johansen cumulative incidence of two competing events for two groups with a number-at-risk table, plus the naive 1 - Kaplan–Meier curve that overstates relapse; the two incidences and overall survival asserted to sum to 1

Techniques: built-in self-check

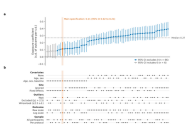


137. Decision-curve analysis

CLINICAL PREDICTION

Decision curves of net benefit against threshold probability for two risk models versus treat-all and treat-none, the winning range shaded, a second axis in false positives per true positive; the treat-all curve asserted equal to its analytic line

Techniques: built-in self-check, shading

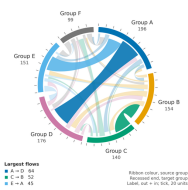


138. Specification curve

STATISTICS

Specification curve of 72 OLS fits (five analytic choices): ranked estimates with 95% CIs, median and main specification marked, above a dot matrix of the choices; hand-coded slope and s.e. asserted equal to linregress and lstsq

Techniques: built-in self-check

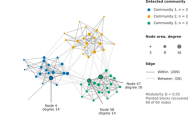


139. Chord diagram

Flows

Chord diagram of a 6×6 directed flow matrix from scratch: arcs spanning each group's total, source-coloured Bézier ribbons, the three largest flows saturated, group labels clear of the tick ring; ribbon ends asserted to tile every arc and the circle to close at 2π

Techniques: direct labels, built-in self-check

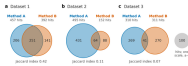


140. Network communities

Networks

60-node stochastic block model drawn with a numpy Fruchterman–Reingold layout, coloured by spectral modularity clustering, node area by degree, within-community edges darker than bridges, hubs labelled; modularity above 0.3 and recovery of the planted blocks asserted

Techniques: direct labels, built-in self-check

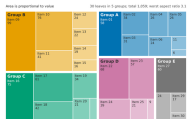


141. Proportional Euler diagrams

Sets

Three area-proportional two-set Euler diagrams on one scale: radii from set sizes, centre distance solved by brentq so the lens equals the intersection, counts placed in each region; lens area asserted equal to the intersection

Techniques: root finding, built-in self-check

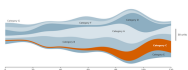


142. Squarified treemap

Hierarchies

Squarified treemap (Bruls–Huizing–van Wijk, coded from scratch) of 30 leaves in 5 groups: group-colour tints, white gutters, labels only where the rendered text fits; one area scale, no overlap and a worst aspect ratio below slice-and-dice asserted

Techniques: direct labels, built-in self-check, aspect-aware rotation

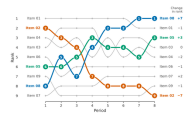


143. Streamgraph

Flows

Streamgraph of 7 categories on the Byron–Wattenberg wiggle baseline, layers ordered inside-out, stroke-free fills with one saturated story layer, labels where each layer is thickest, a scale bar in place of the y axis; layer thickness, column totals and the baseline formula asserted

Techniques: direct labels, built-in self-check, shading

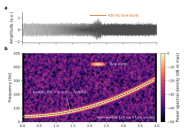


144. Bump chart

Rankings

Bump chart of 9 items over 8 periods: ranks joined by smoothstep connectors, three story items in colour with rank numerals in their markers, the rest grey, names at both ends with net change; ranks asserted to be permutations equal to scipy's rankdata

Techniques: built-in self-check

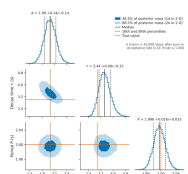


145. Chirp spectrogram

Signal processing

Waveform of a quadratic chirp plus tone burst in noise over its Hann-window STFT spectrogram in dB (rasterized, aligned colorbar), analytic instantaneous frequency overlaid; the spectrogram ridge asserted to follow $f(t)$ within one frequency bin plus window smearing

Techniques: rasterized points, colorbar, built-in self-check

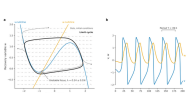


146. Corner plot of a posterior

Bayesian inference

Corner plot of a three-parameter posterior from a hand-coded Metropolis–Hastings sampler (4 chains): step marginals with median and 16th/84th percentiles, 39.3% and 86.5% mass contours, truth crosshairs; R-hat below 1.05, contour mass and truth coverage asserted

Techniques: built-in self-check, contours

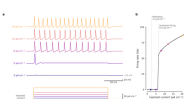


147. Phase portrait

Dynamical systems

FitzHugh–Nagumo phase plane (nullclines, direction field, trajectories onto the limit cycle, unstable focus with its eigenvalues) beside $v(t)$ and $w(t)$ with the period bracketed; fixed point, instability and two independent period estimates asserted to agree

Techniques: built-in self-check

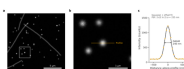


148. Traces with scale bars

ELECTROPHYSIOLOGY

Hodgkin–Huxley responses to five current steps as axis-free stacked traces with an L-shaped scale bar and the step protocol beneath, beside the f–I curve with matching markers; resting fixed point and non-decreasing spike counts asserted

Techniques: built-in self-check

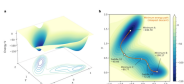


149. Zoom inset and line profile

MICROSCOPY

Simulated bead-and-filament fluorescence field with scale bar and ROI, the ROI magnified pixel-for-pixel with its own bar and profile path, and the line profile with a Gaussian fit and FWHM bracket; fitted FWHM asserted to match the PSF within 5%

Techniques: built-in self-check



150. Energy landscape, 3-D and map

CHEMICAL PHYSICS

Müller–Brown potential as a rasterized 3-D surface over floor contours beside a map with numerically located minima and saddles, their energies and the steepest-descent minimum-energy path; gradients, Hessian signatures and literature minimum energies asserted

Techniques: rasterized points, built-in self-check, contours

3 The failure-mode catalog

Sixteen named patterns, every one harvested from a real bug in building this library. Each surfaced as a wrong *picture* (or a crash) from plausible code; the render-and-inspect loop caught them all.

Escape sequences in raw strings. `\u00b5` inside `r"..."` prints itself. Type the literal character. (Recurred seven times before it became reflex.)

Aspect-blind slope labels. A label rotated by $\arctan(\text{slope})$ ignores the axes' unit or decade aspect and slashes across its own line. Compute the visual angle from the rendered axes geometry (figs. 61, 69, 78; automated in `ashby.py`).

Insets over data. An inset that covers curves must move or become a panel (figs. 53, 63).

Legend–curve collisions. Legends are placed blind; check the swatch column against every curve, or label directly (fig. 70).

Nonlinear secondary axes and minor ticks. A $1/s^2$ transform asks the minor locator to enumerate ticks to 10^{18} ; disable minor locators on nonlinear secondary axes (fig. 77).

Stiff ODEs need terminal events. Boundary-layer shooting blows up past the far boundary; add terminal events instead of praying (fig. 63).

One sign, many symptoms. A single sign slip in one crossing formula truncated a boundary, spawned two stray segments and a bow-tie polygon — four symptoms, one cause (fig. 74). Multiple anomalies usually share a root.

Positional spill into a keyword. `f(ax, *t[1:], label=t[0])` silently feeds a trailing positional into `label`. Unpack tuples explicitly. (Recurred verbatim: figs. 75, 80, 81.)

Fixed-point crawl breaks index arrows. Trajectories crawl near equilibria, so index-based arrow placement collapses to zero length and renders with arbitrary orientation; anchor arrows by *value* and give them a minimum spatial length (fig. 76).

Bottom labels vs. the tick row. Symbolic labels under the axis land on the tick numbers; put them inside the axes (figs. 84, 85).

Converging curves pile their labels. Curves that meet at an edge stack right-edge labels into mush; label each curve where the family is spread (figs. 86, 95).

Guide-line crossover corridors. Two guide lines through different anchors swap vertical order at their crossing; a label corridor that works on one side fails on the other (fig. 80).

Helper contracts are nonlocal. Calling a log–log helper before the axes are logarithmic fed $\log_{10}$ a linear autoscale range, produced NaN, and crashed *inside the font renderer* — errors surface far from their cause (fig. 88).

The renderer speaks a dialect. `\tfrac` is fluent amsL^AT_EX and a fatal error in mathtext (fig. 94). Habits are not portable across renderers.

Arrows point at coordinates, not meanings. The Rayleigh–Jeans annotation landed on the Planck peak instead of the classical curve it named; aim at the named object’s ink (fig. 93).

Algebraic label collisions. Sometimes the collision is a theorem: a label offset exactly equal to the Henderson–Hasselbalch term put every titration label on its own curve (fig. 95); a productivity curve passed through the biomass label because $DX = X$ at $D = 1$ there (fig. 92). When a label lands on a curve at one x , suspect the algebra, then move along the curve.

4 Checklists

Before writing

- Identify the governing equations and the landmark values the figure must honour; plan the built-in self-check.
- Choose column width, panel count and message-per-panel; pick which curves get direct labels and where the family is separated.

Before rendering

- Style sheet loaded; sizes in inches; symbols as literal characters; axis labels carry units; RNG seeded.
- Rotated labels use rendered-aspect angles; secondary nonlinear axes have minor locators disabled; tuples unpacked explicitly.

Inspecting the render

- Every label: on its object, off the data, inside the frame, clear of tick rows and of every other label?
- Every arrow: finite length, pointing at the named ink, correct direction along the curve?
- Landmarks and self-checks: do computed values match the physics (peaks on loci, tangencies exact, areas equal, limits approached)?
- Anything clipped at a spine? Anything only visible because you zoomed? Fix, rerun, look again.

Shipping

- PNG + vector PDF written; confirmation printed with key numbers.
- README line added; library zip rebuilt; script runs standalone from a clean directory.

Closing note. The library’s real product is not the pictures but the loop that made them: derive, render, inspect, fix, and let the physics check itself. Everything else in this guide is that loop, written down.